

Institute of Actuaries of India

Subject CT7 – Business Economics

March 2017 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution: 1.5 Mark to each answer

1. b
2. a
3. b
4. a
5. b
6. a
7. d
8. a
9. b
10. c
11. a
12. c
13. a
14. d
15. c
16. c
17. c
18. d
19. a
20. a
21. c
22. b
23. d
24. c
25. b
26. a
27. d
28. c
29. c
30. d

[Q.No. 1 to 30=45 Marks]

Solution 31:

i) Frequent changes in the currency value could adversely affect trade and investment by creating uncertainty and undermining business confidence. Governments, therefore, may wish to prevent such fluctuations in the currency value. The measures available to the central bank in this regard would depend on whether the aim is to curtail the day-to-day or longer term changes in the exchange rate.

If the government aims to maintain the value of the currency close to a long term equilibrium value, it could buy/sell the domestic currency in the foreign exchange market to prevent it from

falling too far below/above the desired level. If factors causing the downward pressure on the currency cause the demand curve to shift left or the supply curve to shift right, for example, the central bank's intervention will result in reverse shifts in the demand and supply curves and the desired exchange rate will be restored.

To alleviate the downward pressure on the currency, a government could alternatively borrow foreign currency from other countries or international agencies such as IMF. It could use the loan to buy the domestic currency in the foreign exchange market. This will result in returning the demand and supply curves to the original positions and the currency value to resume its original level.

Yet another measure for governments is to raise interest rates temporarily. This will encourage those abroad to deposit their money in the domestic country and the domestic residents to keep their money in their own country. Raising interest rates will result in an increase in the demand for, and decrease in supply of, the currency.

The above measures, though they can be implemented to control day-to-day fluctuations, will not be sustainable in longer term. (Explanation of the reasons in terms of depletion of foreign reserves, high cost of foreign debt, uncertainty about interest rate movements needs to be provided.)

Governments can use fiscal and monetary policies to maintain the value of the currency for longer periods such as months or years. Contractionary fiscal and monetary policies, for example, could be used to dampen aggregate demand. A contractionary fiscal policy would involve raising taxes and /or reducing government expenditure. A contractionary monetary policy would involve raising interest rates to reduce borrowing and hence reduce aggregate demand.

A dampening of aggregate demand will reduce consumer spending including expenditure on imported goods, thereby reducing the supply of the currency in the foreign exchange market. A reduction in aggregate demand will also reduce inflation. With lower prices in the domestic economy, exports will be cheaper and more attractive to consumers abroad who will increase their demand for the domestic currency. There will also be less demand for more expensive foreign goods leading to lower imports and lower supply of the domestic currency. Both effects will result in supporting the value of the currency.

Perhaps the most sustainable measure, for example, to support a currency is for the government to implement supply side policies. This would involve improving the long term competitiveness of the domestic industry by encouraging improvements in the quality of the goods produced and/or reducing production costs. These are achieved by improving the quality of training and/or research and development.

(7)

ii) Governments can restrict access to foreign exchanges which will restrict the outflow of the currency, thereby reducing the supply of the currency to support its value. However, to the extent that long term use of this measure would restrict international trade and free movement of capital, benefits from trade will be foregone. Loss of advantages from international trade are gains from cost differences between countries, decreasing costs resulting from economies of

scale, differences in demand conditions in countries, increased competition, growth in exports in line with growth in world demand and other advantages related to social, political and cultural factors that are foregone. Restricting foreign exchange trading will restrict free movement of capital and inhibits foreign investment in the country. Import restrictions could provoke retaliation by other countries, restricting trade further.

(3)

[10 Marks]**Solution 32:**

Free trade and competition benefits consumers who would have access to global markets and lower prices. Firms gain as technology spreads faster and are therefore able to specialize in activities (products and processes). Policy makers may also find that there is improved political closeness which may help some countries to become more stable and/or bring countries together to resolve differences.

The poor may experience increased inequality and further poverty. This may be because globalization enables multinational corporations (MNCs) which are primarily from wealthy countries to exploit their position in overseas markets. Without local competition, they can pursue profitable activities which may be at the cost of wider social aims. Firms may also use their power to exert pressure on their own governments to develop relations with overseas governments. Hence, it may be viewed as the rich exerting their power upon poorer nations. Moreover, although globalization promotes the sharing of cultural experience, as MNCs spread further and further, the associated cultural influences may become skewed.

[4 Marks]**Solution 33:**

Income measure

This measures the total of factor earnings. These are factor payments to land, labour and capital. Transfer payments are not included (to avoid double counting).

Output measure

This measure the value of goods and services produced. To avoid double counting we only include the "value added" component of firms' output.

Expenditure method

This includes all expenditure (including exports minus imports) in the economy and also what would have to be spent to purchase increases in inventories (or stocks). To avoid double counting, we include only expenditure on final goods.

The three concepts are identically equal. However, in practice difficulties of measurement and collection of data (eg the black economy) mean that the three measures will not be equal.

[4 Marks]**Solution 34:**

i) $\text{Injections} = I + G + X = 550$

$\text{Withdrawals} = S + T + Z = -100 + 0.2Y_d + 0.2Y + 50 + 0.24Y$

$$Y_d = (1 - 0.2)Y = 0.8Y$$

$$550 = -100 + 50 + (0.16 + 0.2 + 0.24)Y$$

$$Y = 1000$$

ie the equilibrium level of national income is 1000.

(2)

ii)

$$a) Z = 0.24Y + 50 = 0.24 \times 1000 + 50 = 290$$

$$b) S = -100 + 0.2Y_d = -100 + 0.2 \times 0.8 \times 1000 = 60$$

$$c) C = Y - I - G - X + Z = 1000 - 200 - 150 - 200 + 290 = 740$$

(3)

iii) 1.67

(1)

[6 Marks]

Solution 35:

i) 30m surplus

(1)

ii) 10m deficit

(1)

iii) -40m

(1)

iv) No, the country does not have a fully flexible exchange rate. We can deduce this from the entry decrease in official reserves" which indicates that the central bank has been purchasing the domestic currency

(1)

[4 Marks]

Solution 36:

i) a) Risk refers a situation in which the probabilities of the different possible outcomes are known, but it is not known which outcome will occur.

(0.5)

b) Uncertainty refers to a situation in which the probabilities of the different possible outcomes are not known.

(0.5)

- ii) 1. holding stocks of goods and services – which can be supplied to the market when prices are favourable
 2. purchasing information, eg market research and trade publications
 3. using futures and forwards
 4. using insurance

(2)

[3 Marks]

Solution 37:

Four marketing strategies would be:

1. Market penetration
2. Product development
3. Market development
4. Diversification

1. Market Penetration:

Selling current product to the current market. The firm gains if the current market expands and people use more than one mobile phone. This is likely to be the least risky strategy. It may lead to increased competition, especially if the market is not expanding. -----

2. Product development: The firm can develop new mobile phones and sell them to the current market. -----

These new product development could involve horizontal or vertical product differentiation. If the company adopts horizontal product differentiation then the mobile phone need to reflect different consumer requirements, but are of similar quality and tend to cost same to produce. The Company will have to do some research work to understand the consumer taste and preference before launching the product.

If the company adopts vertical product differentiation, then the new mobile phones needs to be of better quality than existing one so as to differentiate it from the market. However this may also cost higher to the company.

3. Market development: The firm may conduct research to explore any new market for its existing product. The new market may be in a new physical location, or may be a different segment of the market in the current location.

Possible new location could be expanding to states where the firm has minimal presence currently.

Possible new market segment could be Students if the firm's existing customers largely includes working population. The firm will have to market the product accordingly to make it appealing to this new market segment

4. Diversification: This may involve producing new products for new market. This may be one of the most risky strategy.

Diversification may involve creating similar product or any supplementary product. If the firm currently produces Music player phones then it may explore to produce Business Phones (like blackberry) to appeal the corporate customers.

Also the firm may explore to produce any supplementary products like phone charger, battery, mobile case, audio phones to further expand its product offering and thereby addressing to a larger consumer base.

[9 Marks]

Solution 38:

i) The income elasticity of demand (IED) measures the sensitivity of the quantity demanded to a change in consumer incomes. It is defined as:

$$\% \text{ change in quantity demanded} / \% \text{ change in income} \quad (1)$$

ii) a) The Company X being a monopolistic firm will maximize its profits at output level for which marginal cost is equal to marginal revenue.

The marginal cost is 50 per unit.

To find the marginal revenue function, first find the total revenue function. Total Revenue equals quantity multiplied by price and so the Company X's total revenue function is given by:

$$TR = PQ = 125Q - Q^3$$

Differentiating this gives the marginal revenue function:

$$MR = 125 - 3Q^2$$

Equating this to the marginal cost and solving for Q gives the profit-maximizing output level:

$$125 - 3Q^2 = 50$$

$$\Rightarrow 75 = 3Q^2$$

$$\Rightarrow Q^2 = 25$$

$$\Rightarrow Q^* = 5$$

Substituting this back into the demand curve, gives profit maximising price as:

$$P^* = 125 - Q^2 = 125 - 25^2 = 100 \quad (2)$$

b) Profit = Revenue less Cost

$$\text{Cost} = \text{Variable cost} + \text{fixed cost} = 5 \times 50 + 200 = 450$$

$$\text{Revenue} = \text{Price} \times \text{Quantity} = 100 \times 5 = 500$$

$$\text{Hence, Profit} = 500 - 450 = 50$$

(1)

[4 Marks]

Solution 39:

i) The negative sign means that if price increases, quantity demanded will decrease. The elasticity is - 2, so the quantity demanded will fall by twice the % increase in price.

The price rises from Rs 5 to Rs 5.50, a price rise of 10%, so quantity demanded will fall by 20%.

The quantity demanded of Good X will therefore fall to 320 units. (1)

ii) cross-price elasticity of demand = % change in quantity demanded of Good X / % change in price of Good Y.

The positive sign means that if the price of Good Y decreases, the quantity demanded of Good X will decrease, ie Goods X and Y are substitutes. The cross elasticity is +0.5 so the quantity demanded of Good X will fall by half of the % decrease in the price of Good Y.

The price of Good Y decreases from 6 to 4.5, a price fall of 25%, so the quantity demanded of Good X will fall by 12.5%.

The quantity demanded of Good X will therefore fall to 350 unit. (1)

iii) The income elasticity of demand for Good X is defined as % change in quantity demanded of Good X / % change in income ; which is equal to + 0.6.

Since the income elasticity is +ve, then if income of Mr X increases then quantity demanded would also increase. Quantity demanded would increase by 0.6 of the % increase in Mr X income.

Mr X income increased from Rs 10,000 to Rs 12,000, an increase of 20%. Hence the quantity demanded will increase by 12%.

The quantity demanded of Good X is therefore 448.

(2)

[4 Marks]

Solution 40:

i) If $PED > 1$ (in absolute terms), then an increase in price leads to a larger percentage fall in quantity demanded and so total revenue is reduced.

(0.5)

ii) If $PED < 1$ (in absolute terms), then an increase in price leads to a smaller percentage fall in quantity demanded and so total revenue is increased.

(0.5)

iii) If $PED = 0$, which corresponds to a vertical demand curve, then quantity demanded doesn't change with price, so raising the price by a certain percentage will increase total revenue by the same percentage.

(0.5)

iv) If $PED = -\infty$, which corresponds to a horizontal demand curve, then quantity demanded will increase without limit in response to a small reduction in price, but the firm will lose all its sales if it tries to raise its price.

(0.5)

[2 Marks]

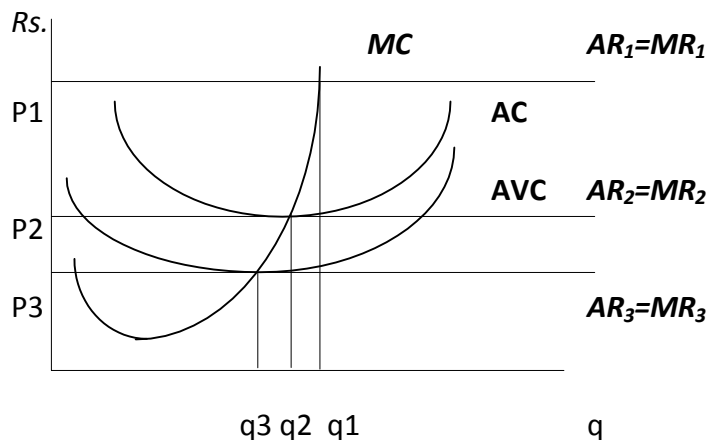
Solution 41:

i) Main assumptions are:

- a. Since each firm has a tiny fraction of the market, all firms are price takers.
- b. There is complete freedom of entry into the industry.
- c. Firms produce an identical (homogeneous) product.
- d. There is perfect knowledge in the market

(2)

ii) A firm's short-run supply curve can be derived by considering how much the firm will produce in the short run at different price levels.



If:

- the price is P_1 , then the firm will produce where $MR = MC$, ie at q_1 – at this price, the firm makes supernormal profit since price exceeds average cost
- the price is P_2 , then the firm will produce where $MR = MC$, ie at q_2 – at this price, the firm makes normal profit since price is equal to average cost
- the price is P_3 , then the firm will produce where $MR = MC$, ie at q_3 – at this price (or any price between P_2 and P_3), the firm makes less than normal profit, but it carries on in the short run because price exceeds AVC
- the price is below P_3 , then the firm will supply nothing, since the price does not even cover its average variable cost.

Therefore, the firm's short-run supply curve is the MC curve above minimum AVC. At prices below minimum AVC, the firm will supply nothing.

(3)

[5 Marks]
